CLASS 7 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 4: **Expressions using Letter-Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The perimeter of a rectangle with sides $7x + 6$ and $2x - 3$ is: 1

(a) $10x + 18$ **(b)** $18x + 6$ **(c)** $28x$ **(d)** $9x + 3$

2. If $n = 14$, then $\frac{n(n+1)}{2}$ equals: 1

(a) 104 **(b)** 106 **(c)** 105 **(d)** 107

3. Which expression is always odd for a whole number n? 1

(a) $4n + 2$ **(b)** $4n + 3$ **(c)** $2n + 4$ **(d)** $n + 3$

4. If $x = 3$ and $y = 6$, the value of $5x - 4y + xy$ is: 1

(a) 11 **(b)** 9 **(c)** 8 **(d)** 10

5. A rule is: output $= 3\times$ input -7. The output for the input 12 is: 1

(a) 28 **(b)** 29 **(c)** 30 **(d)** 31

6. If $x = 5$, the value of $3x^2 + 4$ is: 1

(a) 79 **(b)** 80 **(c)** 81 **(d)** 78

7. If $a = 5$ and $b = 2$, the value of $2a^2 - 9b$ is: 1

(a) 34 **(b)** 32 **(c)** 31 **(d)** 33

8. Simplify $6(x + 1) - 2(x - 4)$: 1

(a) $4x - 2$ **(b)** $8x + 14$ **(c)** $4x + 14$ **(d)** $4x + 2$

9. **Assertion (A):** 3a + 2a = 5a. 1
Reason (R): 3a + 2a = $5a^2$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** 4x + 3y = 7xy. 1
Reason (R): Only like terms can be added to give a single term.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Write the sum of three consecutive multiples of 5, starting with $5n$, and simplify it. 2

12. Find the value of $3x^2 - 5x + 1$ when $x = 4$. 2

13. Simplify: $8(x - 1) - 5(x - 4)$. 2

14. Write the sum of three consecutive multiples of 4, starting with $4n$, and simplify it. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A rectangle has length $(4x + 6)$ cm and breadth $(5x - 2)$ cm. Find its perimeter as an expression and its value for $x = 8$. 3

16. Write the sum of 5 consecutive numbers starting at n as an expression in n, and evaluate it for $n = 6$. 3

17. Taxi A charges ₹40 plus ₹13 per km and taxi B charges ₹100 plus ₹8 per km. Write both fares for k km. For a journey of 10 km, which taxi is cheaper and by how much? 3

OR

A rectangle has length $(6x + 2)$ cm and breadth $(4x - 3)$ cm. Find its perimeter as an expression and its value for $x = 7$. 3

18. Write the sum of 6 consecutive numbers starting at n as an expression in n, and evaluate it for $n = 4$. 3

OR

Taxi A charges ₹25 plus ₹14 per km and taxi B charges ₹65 plus ₹9 per km. Write both fares for k km. For a journey of 9 km, which taxi is cheaper and by how much? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A taxi in Jaipur charges ₹30 as the fixed charge and ₹12 for each kilometre. 5

(i) What is the fare for 5 km?

(ii) Write an expression for the fare for k km.

(iii) Find the fare for 10 km.

(iv) A passenger paid ₹138. How many kilometres did he travel? Show your working.

20. In a pattern, figure n has $3n + 7$ tiles. Find 5

(i) the tiles in figure 30,

(ii) the total tiles in figures 1 and 2, and

(iii) how many more tiles figure 30 has than figure 29.

OR

Priya has ₹x. Her brother has ₹30 more than twice her money. Write expressions for the brother's money and for their total money, and find both when $x = 25$. 5

END OF QUESTION PAPER — Total: 40 Marks